Citation Link Prediction in Scholarly Knowledge Graphs

Hala Abdullah
Department of Computer Science
Prince Sultan University
Email: 222511413@psu.edu.sa

***Abstract*—Scholarly knowledge graphs represent entities such as papers, authors, and citations and the relationships that connect them, but often suffer from insufficient data. This research addresses this limitation by incorporating graph-based machine learning models to extrapolate missing relationships using graphical structural patterns. This evaluation is conducted on KG20C scholarly knowledge graph dataset, a curated benchmark derived from the Microsoft Academic Graph, containing heterogeneous entities and relational triples. The methodology combines supervised machine learning models with knowledge graph embedding models, including TransE and DistMult. Experimental results show that supervised models outperform baseline methods by a margin, with the Neural Network achieving the highest performance with an accuracy of 0.813 and F1-score of 0.810, followed by Random Forest. In contrast, embedding-based models demonstrate limited effectiveness, with TransE having the best MRR, reaching 0.0275 after tuning. These findings highlight how strong graphical structural features are in capturing predictive signals while indicating that embedding-based models require more data diversity and semantic information to perform effectively. The results contribute to improving citation prediction for applications such as academic search, recommendation systems, and research analytics.**

***Index Terms*—knowledge graphs, citation link prediction, graph-based machine learning, knowledge graph embeddings, scholarly data**

I. Introduction

If one is to take a closer look at how scholarly metadata is arranged, it would be discernible that such metadata forms large interconnected networks consisting of research elements such as papers, authors, affiliations, and research domains. In a computational context, these elements can be represented as scholarly knowledge graphs https://arxiv.org/pdf/2307.00865 citation, where nodes represent entities and edges capture relationships such as citation patterns, publication venues, and shared authorship.

However, scholarly knowledge graphs fall victim to missing or incomplete data, which leads to gaps in their relational structure. These gaps result in inadequate capturing dependencies between research elements. To combat this issue, this research focuses on evaluating graph-based machine learning models for citation link prediction in scholarly knowledge graphs. The goal is to deduce missing citation relationships using structural patterns found in the graph.

Solving this problem is important to enhance academic search engines, recommendation systems, and research analytics tools; accurately predicting citation relationships can strengthen information retrieval and knowledge discovery.

The main objectives of this research are to analyze the effectiveness and applicability of graph-based machine learning models for citation link prediction, examine the role relational structure plays in model performance, and evaluate these models using standard knowledge graph metrics.

The research's intended contributions are to provide an analysis on heterogeneous relations in citation prediction, to offer empirical evidence on graph-based machine learning models on real-world scholarly data, and to assist future research on graph machine learning for scholarly knowledge graphs.

The contributions of this work include analyzing heterogeneous relations in citation prediction, providing empirical evaluation of graph-based machine learning models on real-world scholarly data, and supporting future research in graph-based learning for scholarly knowledge graphs.

II. Related Works

Several prior studies are relevant to citation link prediction and scholarly knowledge graphs. Tran and Takasu (2025) [1] introduced the KG20C benchmark for scholarly knowledge graph link prediction and provided baseline evaluation protocols using ranking metrics such as MRR and Hits@k. Their work is highly relevant because it establishes the benchmark dataset used in this project, although it does not focus specifically on citation-edge modeling or relational-context analysis.

Huang et al. (2025) [2] improved general knowledge graph link prediction through multi-chain reasoning and semantic fusion. Their work demonstrates strong results on standard KG benchmarks, but its applicability to curated scholarly citation graphs remains unclear. This motivates further study of graph-based techniques in the scholarly domain.

He et al. (2023) [3] proposed H2CGL for citation impact prediction using hierarchical heterogeneous contrastive graph learning with temporal features. Although this work addresses citation networks, its main task is impact prediction rather than missing-edge prediction, making it related but not directly aligned with the objective of this project.

Rao et al. (2024) [4] explored knowledge graph completion using a pre-trained language model with residual attention and categorical entity information. Their results show the value of

TABLE I
SUMMARY OF RELATED WORKS

Paper	Main Idea	Limitation
Tran & Takasu (2025)	KG20C benchmark for scholarly KG link prediction	No citation-specific modeling analysis
Huang et al. (2025)	Multi-chain reasoning for KG link prediction	Not evaluated on scholarly citation KGs
Rao et al. (2024)	PLM-based KG completion with attention	Depends on textual/type features
He et al. (2023)	H2CGL for citation impact prediction	Different task; temporal focus
Watson & Yong (2024)	Citation recommendation via link prediction	Domain-specific dataset

semantic cues in link prediction, but the method depends heavily on textual and type-based features that may not always be consistently available in curated scholarly knowledge graphs.

Watson and Yong (2024) [5] used link prediction for citation recommendation and ranking through transformer-based graph embeddings and normalized top-frequency features. Their work is relevant because it addresses citation-related ranking, but it is based on a domain-specific corpus rather than a scholarly knowledge graph benchmark.

Overall, these studies show that prior work has addressed benchmark construction, reasoning-based link prediction, temporal citation modeling, semantic-enhanced completion, and citation recommendation. However, there remains a need to examine how structural machine learning methods and baseline knowledge graph embedding models perform specifically on citation link prediction in curated scholarly knowledge graphs. Table I summarizes the main differences among these studies.

III. METHODOLOGY

A. Dataset Description

The project is centered around the KG20C Scholarly Knowledge Graph dataset [1] which was designed and curated to be a heterogenous knowledge graph benchmark dataset that captures structured relationships among multiple entity types within the computer science research field. The dataset was derived from the top twenty computer science conferences from the Microsoft Academic Graph (MAG) and provides comprehensive data on papers, authors, venues, affiliations, domains, and the relationships through citations. KG20C enables link prediction and knowledge graph completion by modeling the entities and interactions as relational triples.

The dataset consists of five textual files with categorical objects as data. 'All_entity_info.txt' and 'all_relation_info.txt' are metadata files that describe the graph schema. The former lists all 16,362 entities with unique identifier, name, and entity type, while the latter contains 5 relational identifiers between the entities. The remaining three files construct the knowledge graph's structure and are preemptively divided into standardized splits: 'train.txt' (48,213 triples), 'valid.txt' (3,670 triples) and 'test.txt' (3,724 triples). Each file consists of (head, relation, tail) triples that represent factual relationships between the scholarly entities [8].

B. Data Preprocessing

To make programming the model easier, the original KG20C text files were converted into .csv format. The three triple files (valid, test, train) were then loaded into separate datasets and an additional column "split" was added in each dataset to explicitly identify the type of partition split it was: validation, testing, or training. Then, the three datasets were concatenated into a single large dataframe with 55607 triples.

Moreover, the dataframe columns were given semantically meaningful renamings: "head_entity", "relation", "tail_entity", "split" instead of ordered positional indices for more clarity and reproducibility, Subsequently, the dataframe was checked for missing values and duplicate triples, all of which were not found. Hence, no cleaning was necessary or required.

C. Categorical Encoding

To ensure uniform representation of entities and relation identifiers, all values in the dataframe were subjected to string formatting, whitespace removal, and lowercase converting. Following that, the values were converted into numerical representations using index-based encoding, and head and tail entities were combined into a single entity to avoid duplicate values, $E = \text{sorted}(H \cup T)$, where H represents the set of head entities and T represents the set of tail entities.

Integer identifiers were assigned through enumeration $\{e : i\}$ and $\{r : i\}$, and each triple was subsequently mapped into numerical form: $h_id = f_E(\text{head_entity})$, $r_id = f_R(\text{relation})$, and $t_id = f_E(\text{tail_entity})$.

D. Labeling and Negative Sampling

To enable supervised learning in the model, a binary label was added to the dataset with a default value of 1, representing valid factual relationships in the knowledge graph. Since knowledge graphs contain only positive triples, negative sample labeling was generated using a randomized corruption sampling strategy [6]., either corrupting the head entity or the tail entity in the triple (head, relation, tail) by replacing the value with another entity in the vocabulary. Consequently, the corrupted triples were assigned a value of 0 to their label.

The negative samples were stored in a new separate dataset and concatenated with the original positively labeled triples. This process resulted in a balanced supervised dataset consisting of 111,214 labeled instances, and a balanced number of positive and negative samples (55,607).

E. Exploratory Data Analysis

The analysis shows an imbalance in the frequency distribution of relation types, which suggests that certain relationships dominate the graph and may bias predictions. The cumulative relation dominance follows a long-tail pattern, where a small subset of relations accounts for a large fraction of triples. For this reason, ranking-based metrics such as MRR and Hits@k are more appropriate than simple accuracy metrics.

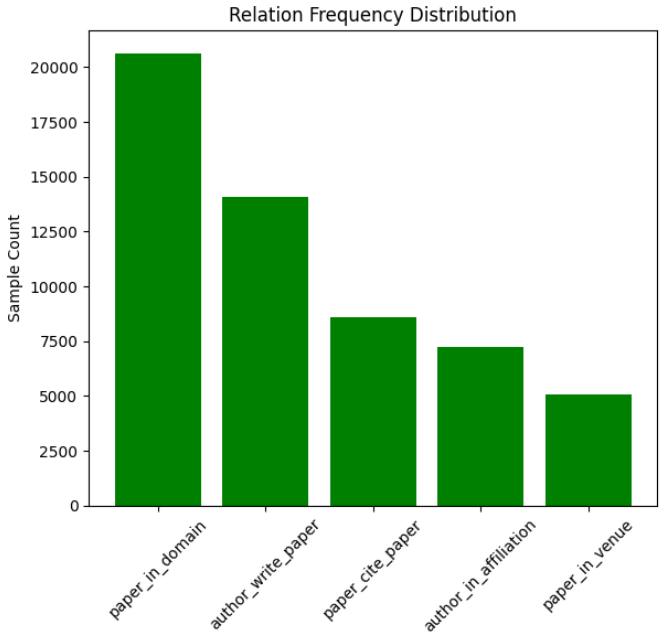

Fig. 1. Relation distribution showing long-tail behavior.

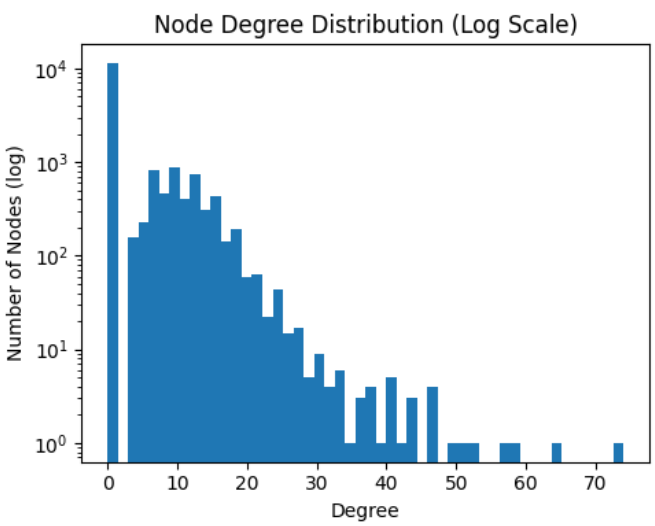

Fig. 2. Entity degree distribution showing hub dominance.

Similarly, the entity degree distribution, computed from both head and tail entities, exhibits a heavy-tail pattern. A small number of entities act as hubs and participate in many triples, while most entities have low connectivity.

These skewed distributions may introduce representation bias and reduce predictability for low-degree entities.

To assess sparsity, the graph density was computed by dividing the number of triples by the square of the number of entities. The resulting value indicates that the KG20C graph is highly sparse, with most entity pairs unconnected. This observation supports the need for negative sampling in supervised learning.

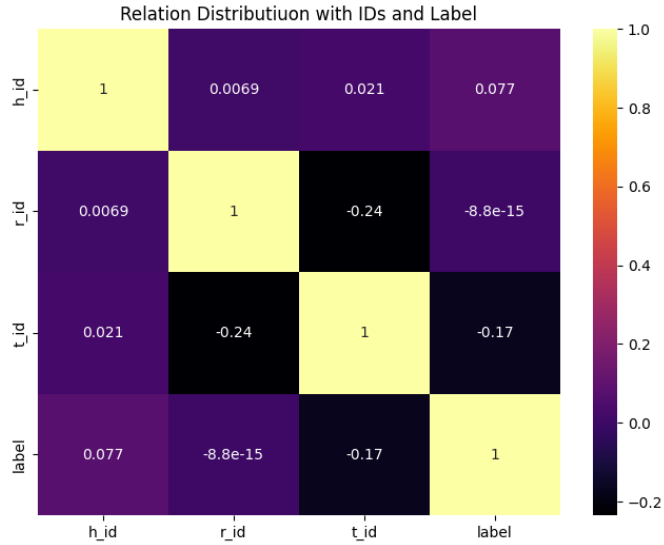

Fig. 3. Correlation heatmap of features and labels.

$$\text{Density} = |T|/(|E|^2) \tag{1}$$

The dataset was also checked for leakage and overlap among the training, validation, and testing splits, and no overlap was found.

The labeled dataset is balanced, with equal numbers of positive and negative samples, ensuring stable optimization and reducing bias caused by class imbalance.

A correlation heatmap confirmed low correlation between identity values and labels, indicating no deterministic bias in the negative sampling process.

F. Feature Engineering

Due to the structural nature of the KG20C dataset observed during exploratory data analysis, feature engineering focused on extracting graph-based statistics.

First, the node degree features head_degree and tail_degree are structural features extracted for each triple(h,r,t), and represent the total number of triples each entity participates in, showing its prominence in the dataset. Furthermore, additional features were derived from the interactions between the two node degree features and represent hub dominance and connectivity asymmetry, including: degree_sum, degree_difference, and degree_product.

Second, two relation-level features are created to capture global relation imbalance and allow the model to learn previous relationship trends without too much reliance on entity identifiers: relation_frequency and relation_probability. Relation_frequency represents the total number of occurences of a relation within the dataset, while relation_probability normalizes the ending value by the total number of triples. The long-tail distribution of relation types shown during EDA, in which a small number of relations predominate the graph structure, is explicitly encoded by the relation features.

Third, conditional interaction features are constructed to model localized relational behavior and measure how often a specific entity appears with a given relation. The feature head_relation_frequency captures the appearance for head entities, and tail_relation_frequency does the same for tail entities. After gaining frequency knowledge for both entities, the counts were normalized by their respective entity degree to help distinguish between globally frequency relations and entity-specific interaction tendencies.

Fourth, logarithmic transformations applied for degree, frequency, and interaction features were formulated via $\log(1+x)$ to stabilize variance, reduce skewness in the heavy-tailed distribution, and improve numerical conditioning.

G. Problem Formulation

To correctly predict whether a candidate triple (h,r,t) represents a valid citation link, the knowledge graph is defined as $G = (E, R, T)$, where E is the entity set, R the relation set, and T the set of true triples.

Two evaluation settings are done: binary classification for ML models to gauge a triple's validity, and ranking-based evaluation for KG models to rank a correct entity among all candidates. The filtered evaluation protocol established by Tran & Takasu (2025) [1] is followed throughout, removing known true triples from the candidate set to avoid penalizing correct alternatives.

1) Simple Baselines: Three simple baselines are implemented to establish lower-bound performance: Random baseline to uniformly predict labels at random, Majority Class baseline to consistently predict the dominant class, and Degree Heuristic baseline to score a triple based on node connectivity.

2) Supervised Structural Machine Learning Models: Supervised structured machine learning models are trained on structural inputs: Logistic Regression as a linear classifier using L2 regularization, Random Forest Classifier as an ensemble of decision trees capturing nonlinear interactions between entities, and Neural Network as a multi-layer architecture with three dense layers (256, 128, and 64), Dropout (0.4, 0.2, 0.2), and a sigmoid output.

Logistic Regression is implemented as a lower-bound linear baseline classifier to evaluate whether citation links can be predicted through linear separability and assess whether simple structual statistics are sufficient for link discrimination. All features were standardized prior to training to guarantee numerical stability, and the model was trained using maximum likelihood estimation with L2 regularization. Meanwhile, Random Forest classifier with 200 decision trees was implemented to model nonlinear interactions and higher-order dependencies such as degree asymmetry and relation strength among structural features.

After that, a multi-layer Neural Network is implemented to model complex nonlinear patterns in the feature space and determine if deep nonlinear transformations provide prediction benefits. The architecture consisted of: Dense(128) layer with ReLU activation and Dropout(0.3), Dense(64) layer with ReLU activation and Dropout(0.2), and an output sigmoid layer, and was trained using binary cross-entropy loss and the Adam optimizer.

3) Knowledge Graph Embedding Models: Knowledge graph embedding models are implemented: TransE [6] and DistMult [7].

Relationships are represented as translations in embedding space by TransE:

$$f(h, r, t) = -\|h + r - t\| \tag{2}$$

After embedding entities and relations into a common vector space, training was performed by head and tail corruption and margin-based ranking loss with negative sampling. By contrast, DistMult computed its ranking score as element-wise multiplication:

$$f(h, r, t) = \langle h, r, t \rangle \tag{3}$$

Through this method, DistMult captures multiplicative entity-relation interactions. Both embedding models are trained using margin-based ranking loss, negative sampling, and Adam optimizer.

H. Evaluation Metrics

To ensure comparability, all models will be evaluated under the same data splits, negative sampling strategy, and ranking protocol. Standard KG link prediction evaluation metrics will be used for the project, including Mean Reciprocal Rank (MRR) Hits@k, and filtered evaluation protocol [8]. MRR is commonly used in KG benchmarks to provide a global measure of how highly correct citation links are ranked by taking the average inverse rank of the link among a list of ranked candidate predictions, and Hits@k measures the proportion of test citation links where the correct entity appears in the top k predicted candidates.

Additionally, the evaluation protocol will include filtered ranking evaluation to avoid penalizing correct alternatives and to ensure realistic evaluation of the model performance. Moreover, the experimental setup will treat citation links as directed edges (i.e. paper $\rightarrow$ cited paper) and evaluation will be performed on the test split, while training split and validation splits will be used for model learning and hyperparameter tuning.

I. Evaluation Protocol Justification

The predefined training (48,213), validation (3,670), and test (3,724) splits in the original KG20C dataset were used to preserve benchmark integrity and ensure comparability with prior studies. The training split was used for model learning, the validation split for tuning, and the test split was reserved for final evaluation.

Because relational dependencies between triples may artificially inflate performance, random re-splitting or k-fold cross-validation was avoided due to the risk of structural leakage in graph data. Leakage checks confirmed that no overlap exists between splits, ensuring unbiased evaluation. After negative sampling, the dataset was converted into a balanced binary classification task.

TABLE II
SUPERVISED MODEL PERFORMANCE AFTER HYPERPARAMETER TUNING

Model	Strategy	Best Configuration	Val AUC
Logistic Regression	Grid Search (5-fold)	$C = 10$, L2 penalty	0.820
Random Forest	Grid Search (2-fold)	$n = 200$, depth=10, leaf=1	0.888
Neural Network	Grid Search + Early Stopping	$lr \times units \times dropout \times batch$	0.888

Given this setup, the appropriate metrics for supervised models were accuracy, precision, recall, and F1-score, while ranking-based metrics such as MRR and Hits@k remained relevant because of the structural sparsity and relation imbalance in the graph.

J. Hyperparameter Tuning

Hyperparameter tuning was conducted to ensure fair and unbiased comparison across all models. For supervised models, tuning was performed using grid search, cross-validation, and ROC-AUC as the objective metric. For embedding models, hyperparameters were tuned using validation-based MRR optimization.

TransE was tuned over embedding dimension, learning rate, and margin, while DistMult was tuned over embedding dimension and learning rate. The best configurations were then retrained to obtain final results.

A grid search over the regularization strength C showed that moderate regularization yields the best performance. Extremely small values lead to underfitting, while larger values provide no significant improvement. Performance remained relatively stable across settings, indicating that the feature space is already well-conditioned and approximately linearly separable.

The Neural Network tuning heatmap shows that the learning rate has a greater influence on validation performance than network depth. Increasing model size produces only slight improvement, suggesting that most predictive signal is already captured by the structural features and that additional architectural complexity yields diminishing returns.

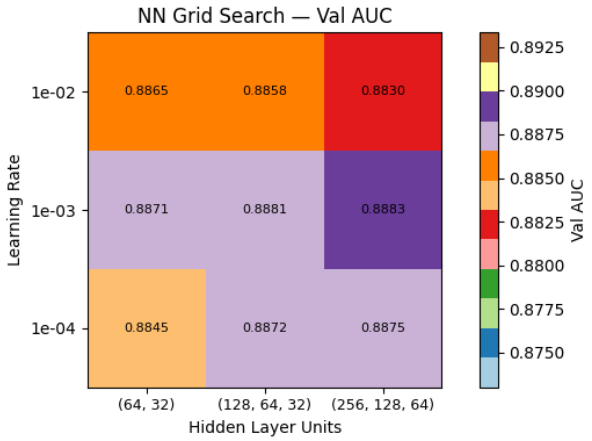

Fig. 4. Tuned Neural Network heatmap.

TABLE III
EMBEDDING MODELS AFTER HYPERPARAMETER TUNING

Model	Best Dim	Best LR	Best Margin	Combos	Val MRR
TransE	100	3×10^{-4}	2.0	27	0.0551
DistMult	100	1×10^{-3}	N/A	9	0.0196

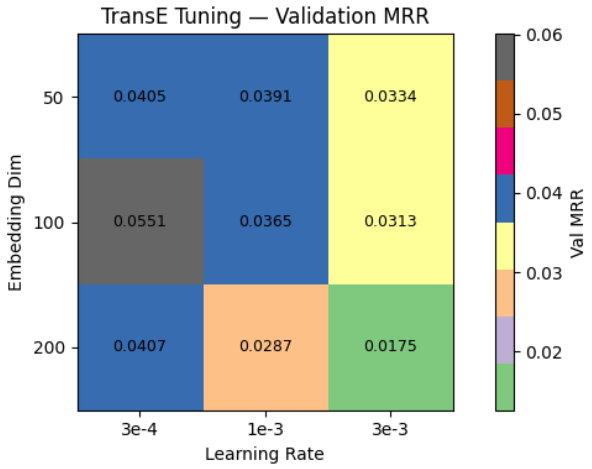

Fig. 5. Tuned TransE heatmap.

The TransE tuning heatmap shows that a modest learning rate (3×10^{-4}) consistently produces the best validation MRR across embedding dimensions, suggesting more stable optimization. Performance deteriorates with larger embeddings, either due to overfitting or more difficult tuning.

IV. RESULTS AND DISCUSSION

A. Supervised Structured Machine Learning Models

All three tuned machine learning models outperform the simple baselines, confirming that graph-structural features carry genuine predictive signals for citation link prediction. The degree heuristic, which scores triples solely by node connectivity, achieves AUC = 0.642, showing that raw node

TABLE IV
SUPERVISED MACHINE LEARNING MODELS PERFORMANCE

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Random Baseline	0.5015	–	0.5015	0.5015	0.5000
Majority Class	0.5000	–	0.5000	0.3333	N/A
Degree Heuristic	0.6094	–	–	0.6093	0.6422
Logistic Regression (tuned)	0.7427	0.7427	0.7427	0.7427	0.8181
Random Forest (tuned)	0.8142	0.8316	0.8142	0.8118	0.8867
Neural Network (tuned)	0.8134	0.8341	0.8134	0.8105	0.8878

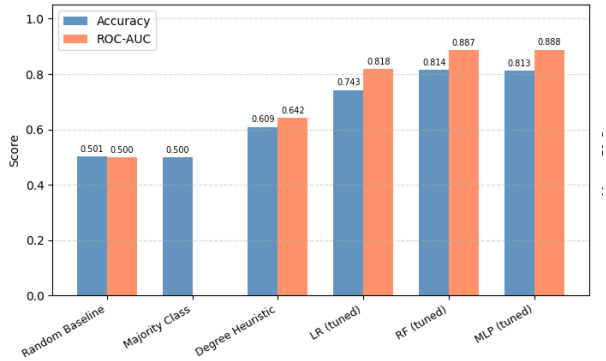

Fig. 6. Model comparison using Accuracy and ROC-AUC.

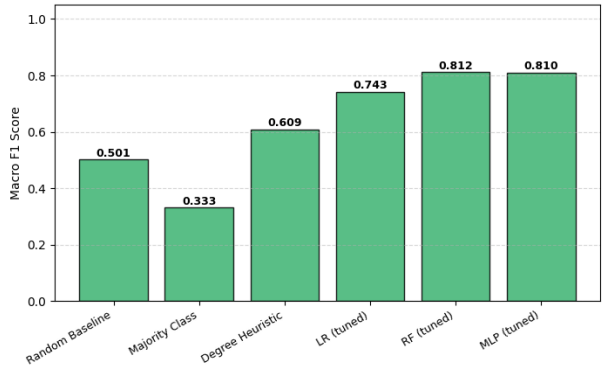

Fig. 7. Macro F1-score comparison across models.

popularity alone is insufficient and that richer structural features are necessary for discrimination. The Neural Network achieves the highest AUC of 0.888, marginally outperforming Random Forest. Logistic Regression, while competitive at AUC = 0.818, confirms that log-transformed degree features create an approximately linear decision boundary.

As shown in Fig. 6 and Fig. 7, Random Forest and Neural Network perform similarly, confirming their equivalence as the strongest classifiers in this task.

B. Feature Importance Analysis

Tail-relation interaction features are the most important predictors in the feature importance analysis, with log-transformed tail-relation frequency having the highest relevance. This suggests that the frequency with which a tail entity appears with a given relation strongly influences the likelihood of a citation. Localized relational patterns appear

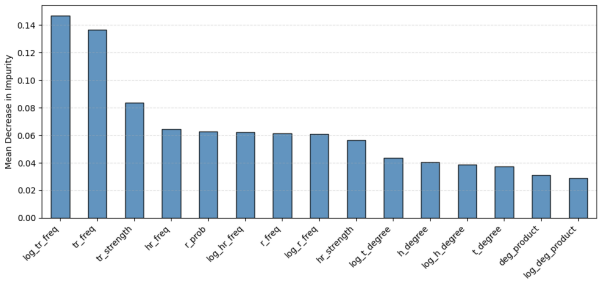

Fig. 8. Top 15 feature importance scores from Random Forest.

TABLE V
KNOWLEDGE GRAPH EMBEDDING MODELS PERFORMANCE

Model	MRR	Hits@1	Hits@3	Hits@10
TransE	0.0186	0.0064	0.0170	0.0405
DistMult	0.0196	0.0102	0.0198	0.0358
TransE (tuned)	0.0275	0.0111	0.0267	0.0591
DistMult (tuned)	0.0165	—	—	0.0301

TABLE VI
BENCHMARK COMPARISON WITH EMBEDDING MODELS

Model	MRR	Hits@10
TransE (tuned)	0.0275	0.0591
DistMult (tuned)	0.0165	0.0301
CPh	0.215	0.348
MEI	0.230	0.368
Random	0.001	$< 5 \times 10^{-4}$
Word2Vec	0.068	0.177

to be more informative than global connectedness, since interaction features frequently outperform simple node degree measures. Hub dominance alone does not fully explain citation behavior, while relation-frequency features contribute more meaningfully to prediction performance [8].

C. Knowledge Graph Embedding Models

The results show that all Hits@k measures increase consistently after hyperparameter tuning, suggesting improved ranking quality. TransE benefits substantially from tuning, improving from MRR = 0.0186 to 0.0275 (+48%). This improvement is associated with a lower learning rate and an optimal embedding dimension, which stabilize training and improve separation between positive and negative triples.

In contrast, DistMult does not show comparable gains and declines slightly in both MRR and Hits@10. Its symmetric scoring function limits its ability to represent asymmetric citation relationships [7].

Overall, both TransE and DistMult serve as useful baseline models for sparse scholarly knowledge graphs. However, their limited performance highlights the difficulty of link prediction in sparse academic graph settings.

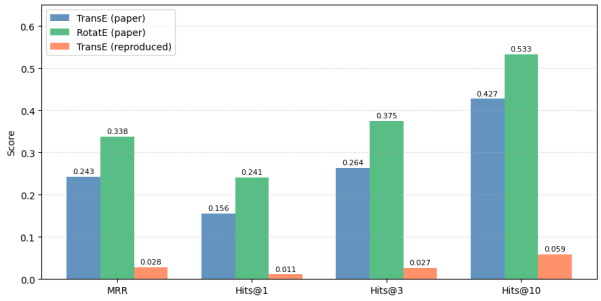

Fig. 9. TransE compared with reproduced and literature baselines.

D. Benchmark Comparison Results

Tran & Takasu (2025) [1] benchmark CPh and MEI, not TransE or DistMult directly. TransE [6] and DistMult [7] are reproduced in this work as additional baselines, with CPh and MEI serving as the primary upper-bound reference targets.

The benchmark comparison shows a substantial performance gap between the reproduced embedding models and stronger literature baselines. Compared with CPh and MEI, which achieve MRR values above 0.20, tuned TransE reaches only 0.0275 and Hits@10 of 0.0591. Nevertheless, the reproduced models still outperform the random baseline, showing that they capture meaningful structural patterns. TransE also outperforms DistMult, supporting the advantage of translation-based models for asymmetric citation relationships.

The comparison in Fig. 9 further illustrates this gap. Although shallow embedding methods such as Word2Vec perform slightly better than random guessing, they remain far below stronger approaches such as CPh and MEI. These results indicate that more expressive models, richer feature representations, and more sophisticated optimization strategies are needed to achieve competitive performance on KG20C.

E. Discussion

The results show that citation link prediction in KG20C depends strongly on graph-structural properties, especially those that capture localized relational behavior between entities and relation types. In particular, tail-relation frequency features have a substantial influence on prediction quality, indicating that stable relational interaction patterns are more informative than node degree alone [8].

The near-parity between Random Forest (AUC = 0.887) and Neural Network (AUC = 0.888) suggests that the available structural features already capture most of the predictive signal. This indicates diminishing returns from increasing model complexity and suggests that further improvements may require incorporating semantic information, such as entity descriptions or textual context, rather than relying only on graph structure [4].

The Random Forest error analysis reveals additional insights. Out of 22,243 test triples, 4,132 were misclassified, yielding an overall error rate of 18.58%. False positives account for 15.00% of cases, while false negatives account for 3.57%, suggesting that the model tends to overpredict the existence of relationships. This behavior is consistent with bias introduced by high-frequency nodes and hub structures in complex networks [9].

Error rates also vary across relations. Relation r_id=1 shows the highest error rate (26.77%), followed by r_id=0 (20.50%) and r_id=2 (17.23%). In contrast, relation r_id=3, which has greater support, yields a lower error rate (14.24%). This suggests that frequent relational patterns are easier to predict, while less common relations remain more difficult [1].

Finally, the benchmark comparison highlights the gap between reproduced models and stronger methods such as CPh and MEI. While the implemented models outperform the random baseline, they remain far below state-of-the-art performance. This gap likely reflects differences in model expressiveness, training strategy, and the absence of semantic or contextual information. Likewise, TransE and DistMult remain limited in sparse scholarly graphs, with DistMult particularly constrained by its symmetric scoring function [6], [7].

V. Limitations and Future Work

Several limitations remain despite strong performances from the machine learning models.

First, the research notably relies on graph-structural features without taking semantic information into consideration, such as: textual descriptions of papers and authors. This negligence potentially restricts representational richness. Second, the KG20C dataset used for building the models consists of scholarly data from a specific set of conferences, which limits its generalizability and scalability when applied to other academic domains. Third, model robustness is likely affected by the random negative sampling strategy, as it does not reflect realistic hard negative cases. Fourth and finally, graph embedding models such as TransE and DistMult show limited performance on the KG20C dataset, which indicates that these models are insufficient to capture complex relationships in sparse scholarly knowledge graphs.

Future work can address these limitations in several ways. First, incorporating semantic information, such as textual embeddings of papers and authors, may improve prediction performance by capturing contextual relationships beyond graph structure. Second, more expressive models, including graph neural networks and advanced knowledge graph embedding approaches, could be explored to better model heterogeneous and asymmetric relationships observed in citation graphs. Finally, extending the evaluation to more diverse scholarly datasets would allow for a more comprehensive assessment of model robustness and generalizability across domains.

VI. Conclusion

This research evaluated graph-based machine learning methods for citation link prediction in the KG20C scholarly knowledge graph. The results show that supervised machine learning models significantly outperform baseline models, with the Neural Network achieving the highest performance (AUC = 0.888), closely followed by Random Forest (0.818).

In contrast, knowledge graph embedding models such as TransE and DistMult demonstrate limited performance, with the best MRR reaching only 0.0275, highlighting the difficulty of modeling sparse and complex relational structures using simple embedding approaches.

These findings indicate that graph-structural features capture strong predictive signals for citation link prediction, while embedding-based methods require more expressive architectures to achieve competitive performance. Embedding-based models could improve if semantic information is incorporated. Overall, the study demonstrates the importance of combining structural information with more advanced modeling techniques to improve link prediction in scholarly knowledge graphs.

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